

Maths-1 - Functions Work-along Sheet

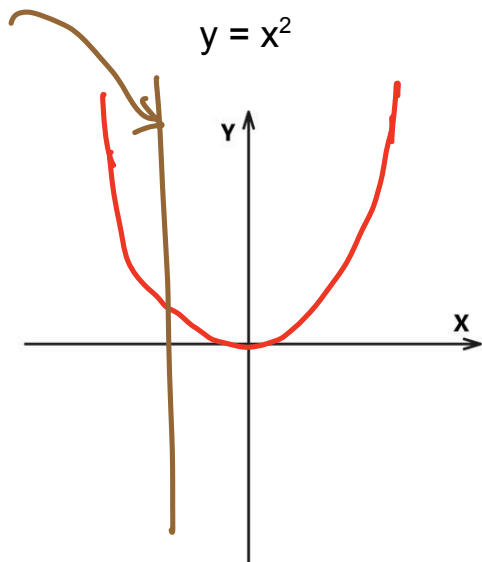
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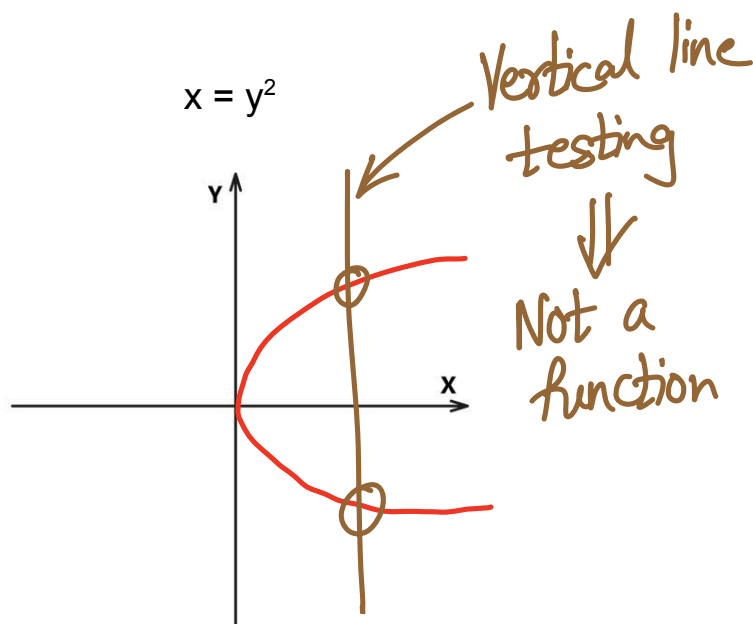
A function is a rule that assigns each element of a set called the domain to exactly one element of another set called the co-domain.

A function is a relationship between variables where each input corresponds to exactly one output(s).

Vertical
line
testing
↓
Function



x	y
-2	4
-1	1
0	0
1	1
2	4



Vertical line
testing
↓
Not a
function

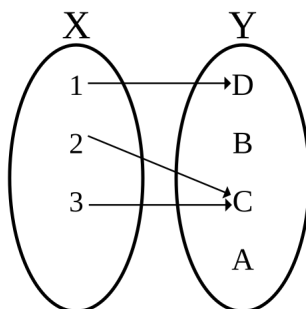
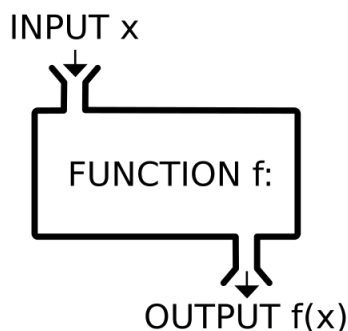
x	y
4	-2
1	-1
0	0
1	1
4	2

In case of $x = y^2$, we get two different y values for the same x value. After looking at the relationship between x and y, we can conclude that $y = x^2$ is a function & $x = y^2$ is not a function.

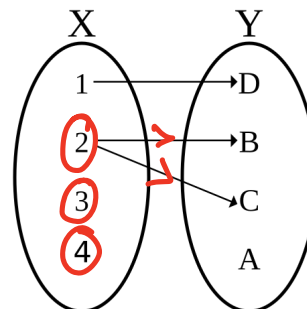
Vertical line test: If any vertical line touches a graph more than once, the graph is not a function.

Classical definition: A function is a rule assigning one number to another.

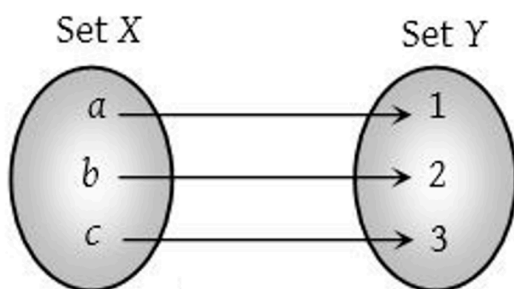
Modern definition: A function is a rule assigning an element of one set to an element of another.



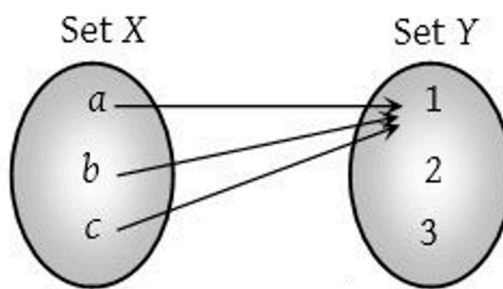
Function



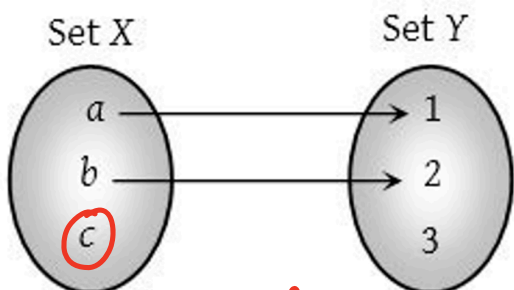
Not a function



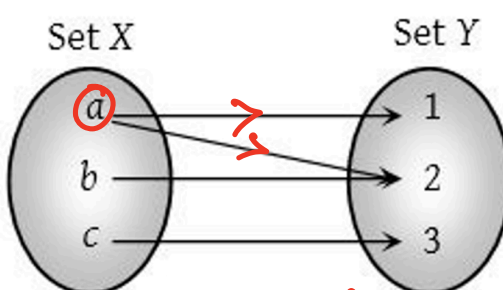
Function



Function



Not a function



Not a function

Valid function rule: Each input maps to exactly one output.

Declaration of function:

$$f: X \rightarrow Y$$

Name of function : Domain of function $\rightarrow$ Co-domain of function

A function from set X to set Y , written as $f: X \rightarrow Y$, is a rule that assigns to each element of $x \in X$ one element $y \in Y$. This assignment is denoted by $y = f(x)$.

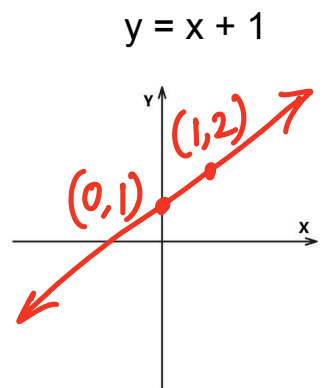
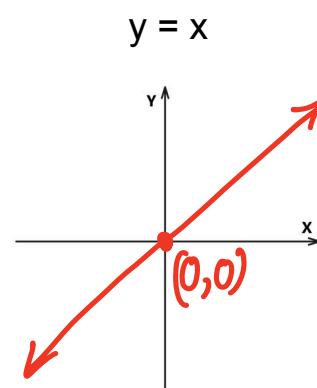
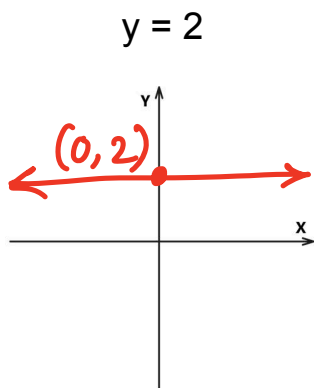
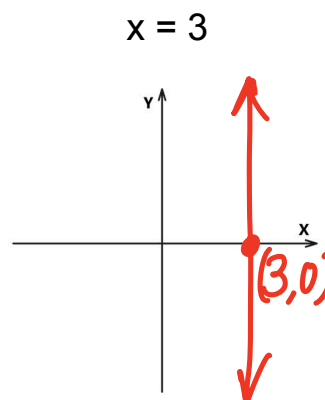
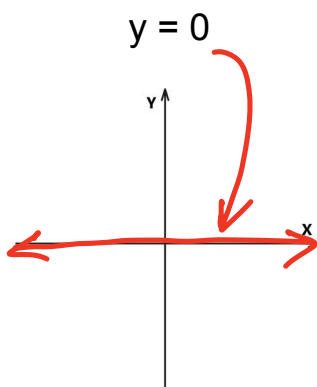
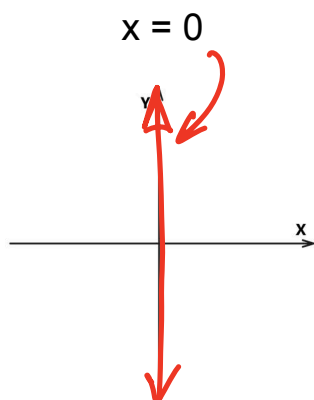
Definition of function:

$$y = f(x)$$

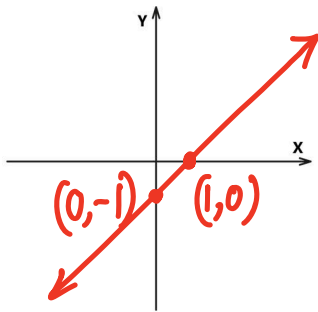
x is the independent variable (input), y is the dependent variable (output).

The set of all allowable inputs is called domain. The set of all potential outputs (not necessarily all will be used) is called co-domain. The set of outputs actually produced by the function (subset of the co-domain) is called range.

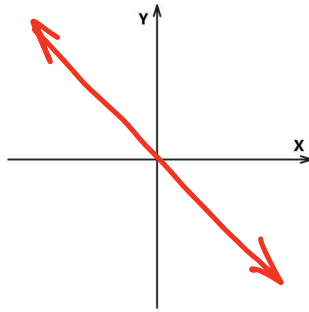
Now, let's get comfortable with plotting graphs initially.



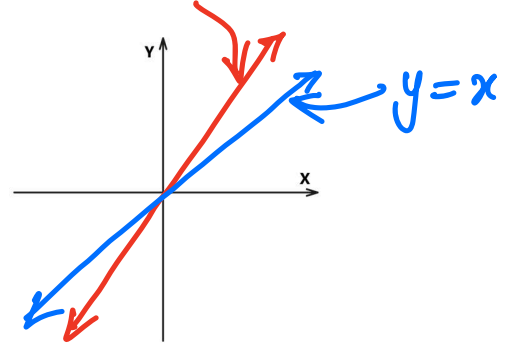
$$y = x - 1$$



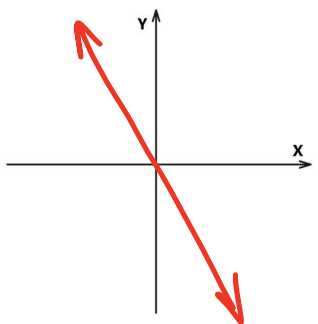
$$y = -x$$



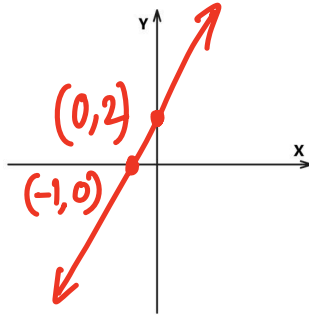
$$y = 2x$$



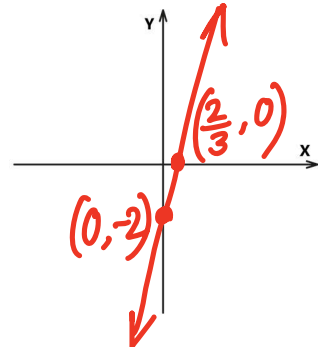
$$y = -2x$$



$$y = 2x + 2$$

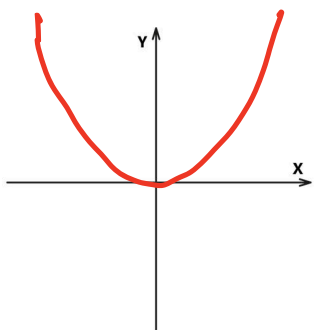


$$y = 3x - 2$$

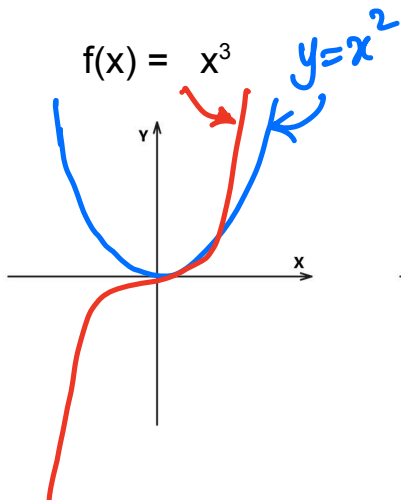


The equation of a line is $y = mx + c$, where m is the slope and c is the y-intercept. You will use this logic of changing the slope and y-intercept when you learn about the Gradient Descent algorithm in Machine Learning.

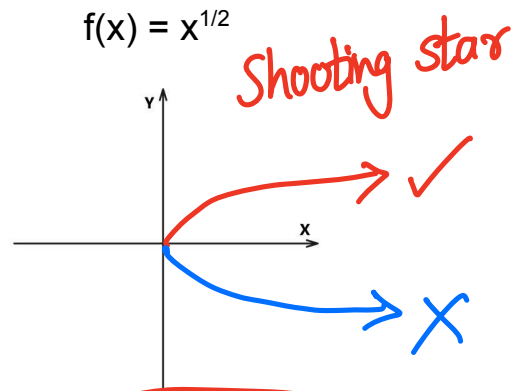
$$f(x) = x^2$$



$$f(x) = x^3$$



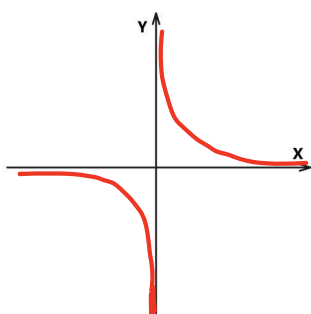
$$f(x) = x^{1/2}$$



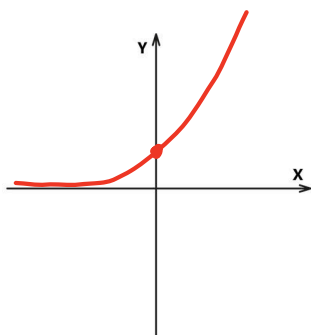
$$\sqrt{4} = \pm 2$$

Why we took only positive values?

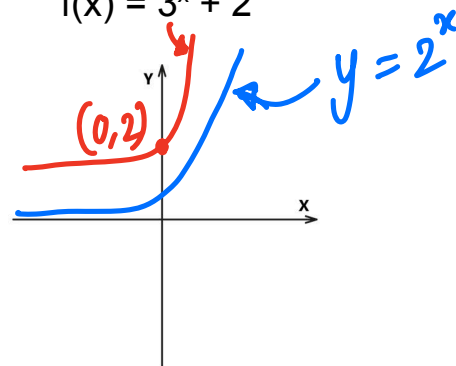
$$f(x) = 1/x$$



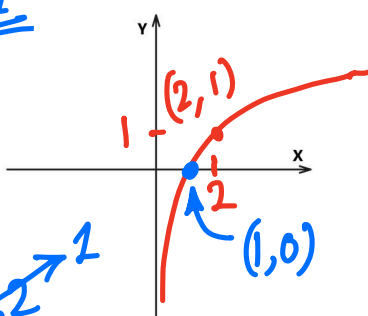
$$f(x) = 2^x$$



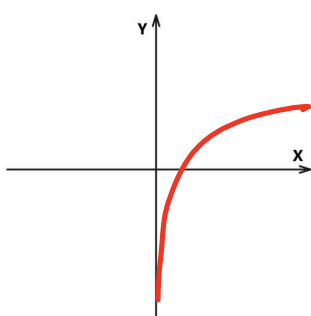
$$f(x) = 3^x + 2$$



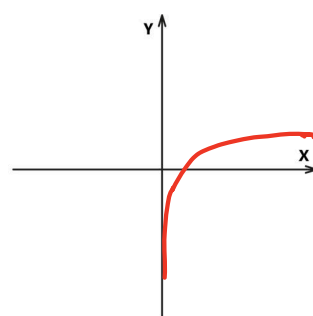
$$f(x) = \log_2(x)$$



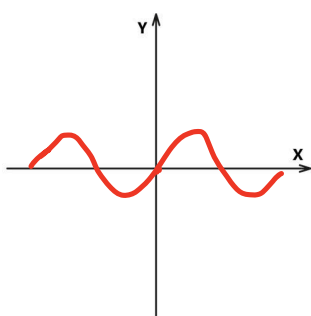
$$f(x) = \log_e(x)$$



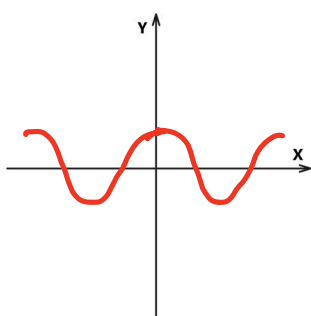
$$f(x) = \log_{10}(x)$$



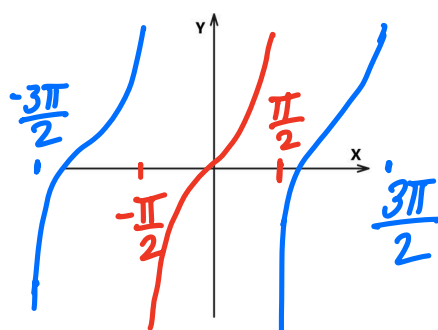
$$f(x) = \sin(x)$$



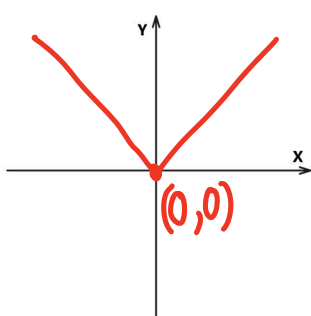
$$f(x) = \cos(x)$$



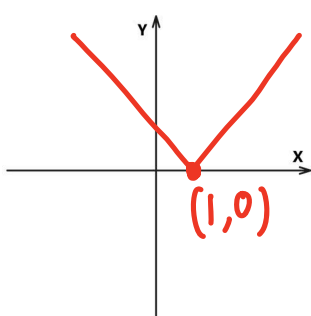
$$f(x) = \tan(x)$$



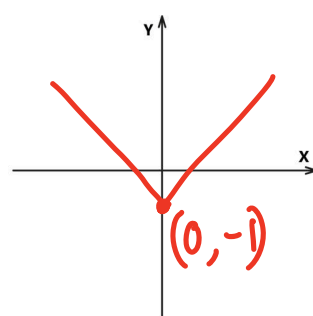
$$f(x) = |x|$$



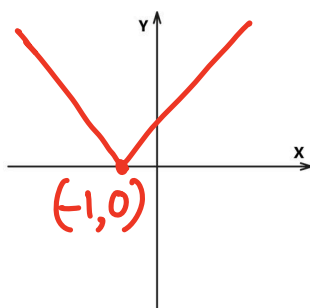
$$f(x) = |x - 1|$$



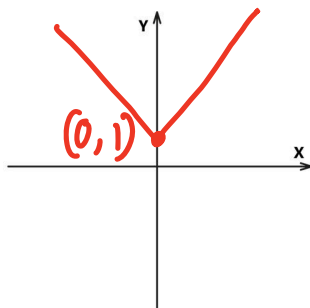
$$f(x) = |x| - 1$$



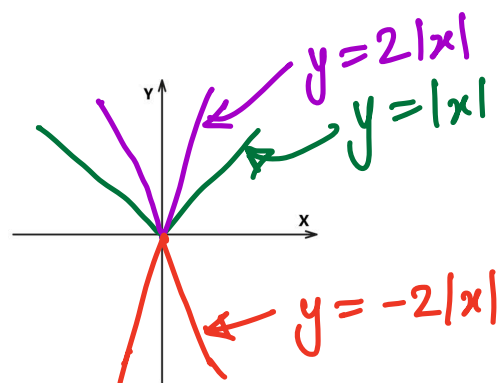
$$f(x) = |x + 1|$$



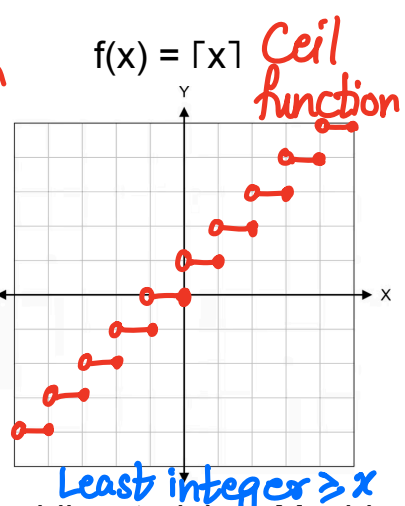
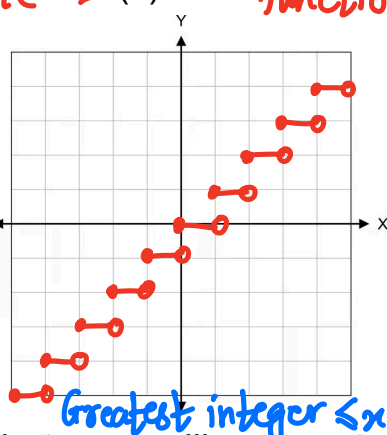
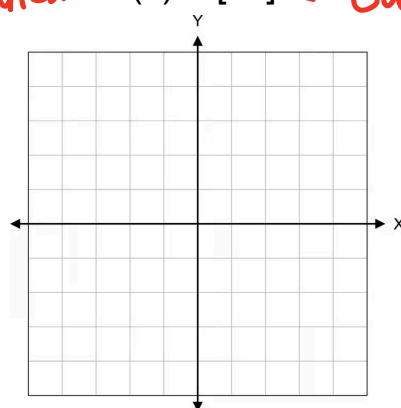
$$f(x) = |x| + 1$$



$$f(x) = -2|x|$$



Step function $f(x) = [x]$ ← Same → $f(x) = \lfloor x \rfloor$ Floor function



Some useful functions that you will encounter while studying Machine Learning are as follows:

Sigmoid	Tanh	RELU
$g(z) = \frac{1}{1 + e^{-z}}$	$g(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}}$	$g(z) = \max(0, z)$

Q. Find the domain, co-domain, and range of $f(x) = 2x + 3$

- $f(x)=2x+3$ is a linear function.
- **Domain:** All real numbers ($\mathbb{R}$) since any real x can be substituted.
- **Co-domain:** Usually taken as real numbers ($\mathbb{R}$)
- **Range:** All real numbers ($\mathbb{R}$), since for every real output y , we can find x = $(y-3)$ / 2 .

Q. Find the domain, co-domain, and range of $f(x) = \sqrt{x-1}$

- The expression inside the square root must be greater than or equal to zero. $x-1 \geq 0 \Rightarrow \underline{\underline{x \geq 1}}$
- Domain = $[1, \infty)$
- Co-domain = $\mathbb{R}$ (default)
- Range = $[0, \infty)$, since the square root always gives non-negative values.

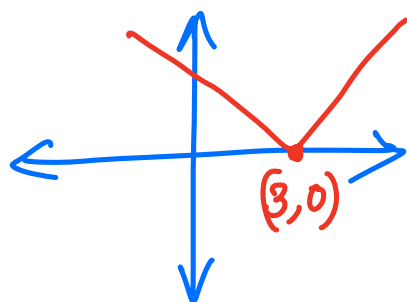
Q. Find the domain, co-domain, and range of $f(x) = \frac{1}{x-2}$

- Denominator cannot be zero
- $x-2 \neq 0$ $\Rightarrow$ $x \neq 2$
- Domain = $\mathbb{R} - \{2\}$
- Co-domain = $\mathbb{R}$
- $x = (\frac{1}{y}) + 2$. Therefore, y cannot be 0 .
- Range = $\mathbb{R} - \{0\}$

Q. Find the domain and range of $f(x) = \frac{x^2 - 4}{x - 2}$

Homework

Q. Find the domain and range of $f(x) = |x - 3|$



$$\text{Domain} = \mathbb{R}$$

$$\text{Range} = [0, \infty)$$

Q. Find the domain and range of $f(x) = \frac{1}{\sqrt{x-1}}$

$$\sqrt{x-1} > 0$$

$$\therefore x-1 > 0$$

$$\therefore \underline{\underline{x > 1}}$$

$$\text{Domain} = (1, \infty)$$

$$\text{As } x \rightarrow 1^+, f(x) \rightarrow \infty$$

$$\text{As } x \rightarrow \infty, f(x) \rightarrow 0^+$$

$$\text{Range} = (0, \infty)$$

Q. Find the domain and range of $f(x) = \sin x + 2$

$$\text{Domain of } \sin x = \mathbb{R}$$

$$\text{Range of } \sin x = [-1, 1]$$

$$\text{Domain of } \sin x + 2 = \mathbb{R}$$

$$\text{Range of } \sin x + 2 = [-1+2, 1+2] = [1, 3]$$

Q. Find the domain and range of $f(x) = \log(x - 4)$

Argument of logarithm must be positive.

$$\therefore x - 4 > 0$$

$$\therefore \underline{x > 4}$$

$$\text{Domain} = (4, \infty)$$

$$\text{Range} = \mathbb{R}$$

Q. Find the domain and range of $f(x) = \frac{1-x}{1+x}$

$$1+x \neq 0$$

$$\therefore \underline{x \neq -1}$$

$$\therefore \text{Domain} = \mathbb{R} - \{-1\}$$

$$(-\infty, -1) \cup (-1, \infty)$$

$$y = \frac{1-x}{1+x}$$

$$\therefore y(1+x) = 1-x$$

$$\therefore y + yx = 1-x$$

$$\therefore yx + x = 1-y$$

$$\therefore x(y+1) = 1-y$$

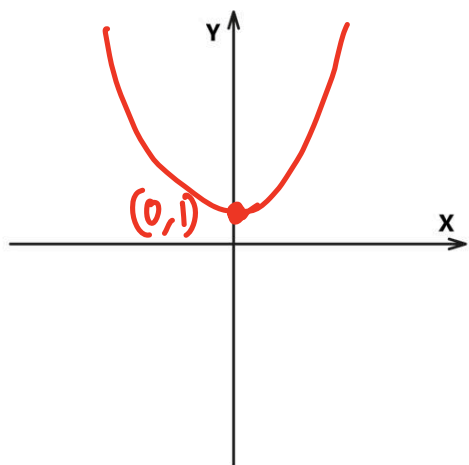
$$\therefore x = \frac{1-y}{y+1}$$

$$y+1 \neq 0$$

$$\therefore \underline{y \neq -1}$$

$$\therefore \text{Range} = \mathbb{R} - \{-1\}$$

Q. Find the domain and range of $y = f(x) = x^2 + 1$



Domain focuses on x values.

Range focuses on y values.

The domain of the given function is

$$\underline{\{x \in \mathbb{R}\}}$$

i.e., $\underline{(-\infty, \infty)}$

The range of the given function is

$$\underline{\{y \in \mathbb{R} \mid y \geq 1\}}$$

i.e., $\underline{[1, \infty)}$

What is the domain of the function $y = \sqrt{1 - x^2}$?

$$\begin{aligned} 1 - x^2 &\geq 0 \\ \therefore 1 &\geq x^2 \\ \text{i.e., } x^2 &\leq 1 \Rightarrow \underline{\underline{-1 \leq x \leq 1}} \\ \therefore \text{Domain} &= [-1, 1] \end{aligned}$$

What is the domain of the function $y = \sqrt{x - 1}$?

$$\begin{aligned} x - 1 &\geq 0 \\ \therefore x &\geq 1 \\ \underline{\underline{x \geq 1}} \\ \therefore \text{Domain} &= [1, \infty) \end{aligned}$$

What is the domain of the function $y = \sqrt{2 - x}$?

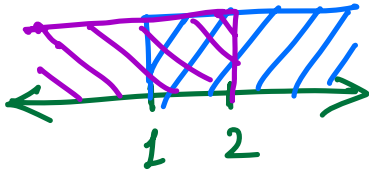
$$\begin{aligned} 2 - x &\geq 0 \\ \therefore 2 &\geq x \\ \text{i.e., } x &\leq 2 \\ \underline{\underline{x \leq 2}} \\ \therefore \text{Domain} &= (-\infty, 2] \end{aligned}$$

What is the domain of the function $y = \sqrt{x-1} + \sqrt{2-x}$?

$$\text{Domain} = [1, \infty)$$

$$\text{Domain} = (-\infty, 2]$$

$$\text{Domain} = \underline{\underline{[1, 2]}}$$



Now, try to find the domain of the following:

$$y = \sqrt{x-2} + \sqrt{1-x}$$

$$x-2 \geq 0$$

$$\underline{\underline{x \geq 2}}$$

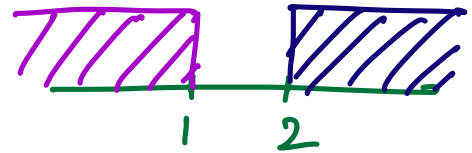
$$\text{Domain} = [2, \infty)$$

$$1-x \geq 0$$

$$\therefore 1 \geq x$$

$$\text{i.e., } \underline{\underline{x \leq 1}}$$

$$\text{Domain} = (-\infty, 1]$$



$$(-\infty, 1] \cap [2, \infty) = \emptyset$$



It is not a function.

It is called formula (or analytical expression)

Note: Function is an assignment of element from a set to elements of other set with conditions (rules) whereas formula (or analytical expression) is any expression relating x with y .

→ Not every formula defines a function.

Function $\equiv$ Formula + "something extra"

→ More examples of formula which are not functions:

$$y = \frac{1}{\sqrt{x}} + \frac{1}{\sqrt{-x}}$$

$$y = \log(x) + \log(-x)$$

$$y = \log(\sin x - 2)$$

→ Not all functions can be written as formula.

Example: Dirichlet Function (Devil's Function)

$$y = \begin{cases} 1 & ; x \in \mathbb{Q} \\ 0 & ; x \notin \mathbb{Q} \end{cases}$$

This is well-defined function but do not have any analytical expression or formula.

→ Functions should be identified with their formula.

Q.
$$y = \begin{cases} \cos x & ; x < 0 \\ 1+x^2 & ; 0 \leq x \leq 2 \\ \log(x-1) & ; x > 2 \end{cases}$$

How many functions are defined above?

→ ONLY ONE

$$\left. \begin{array}{l} y = \cos x \\ y = 1+x^2 \\ y = \log(x-1) \end{array} \right\}$$

These are 3 formulas which come together to make 1 function.

Q.
$$\begin{array}{l} y = \cos x \\ [0, \pi] \rightarrow \mathbb{R} \end{array}$$

$$\begin{array}{l} y = \cos x \\ [\pi, 3\pi] \rightarrow \mathbb{R} \end{array}$$

Are these two same or different functions?

→ Different functions because domains are different.